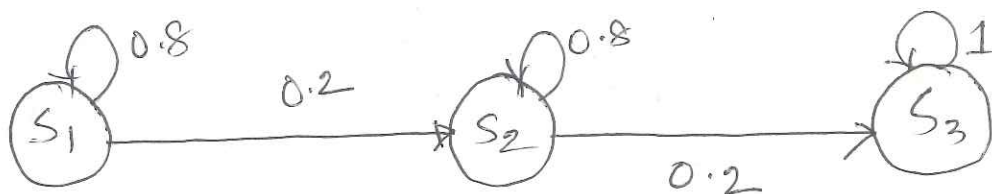


Q. The structure of Hidden states:



transition probabilities

$$a_{ij} = \begin{matrix} & \begin{matrix} s_1 & s_2 & s_3 \end{matrix} \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \end{matrix} & \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0 & 0.8 & 0.2 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Observation numbers of island in the vertical slice

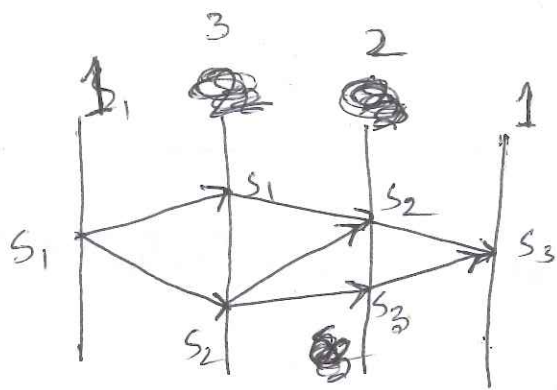
for character A $b_{jk} = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0.9 & 0.1 & 0 \end{bmatrix}$ ~~A~~

for character B, $b_{jk} = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0 & 0.2 & 0.8 \\ 0.6 & 0.4 & 0 \end{bmatrix}$ ~~B~~

Suppose that after character image segmentation the following sequence of island numbers in 4 slices was observed. $\{1, 3, 2, 1\}$

What HMM is more likely to generate this observation sequence, HMM for 'A' or HMM for 'B'?

Ans: HMM for character A:



Output probabilities for character A.

Output probability s_1, s_2, s_3

$$= P(1/s_1) * P(s_1/s_1) * P(3/s_1) * P(s_2/s_1) * P(2/s_2) \\ * P(s_3/s_2) * P(1/s_3)$$

$$= b_{11} * 0.8 * b_{31} * 0.2 * b_{22} * 0.2 * b_{31}$$

$$= 0.9 * 0.8 * 0 * 0.2 * 0.8 * 0.2 * 0.9$$

$$= 0$$

Output probability $s_1 s_2 s_2 s_3$

$$= P(1/s_1) * P(s_2/s_1) * P(3/s_2) * P(s_2/s_2) * P(2/s_2) \\ * P(s_3/s_2) * P(1/s_3)$$

$$= b_{11} * 0.2 * b_{23} * 0.8 * b_{22} * 0.2 * b_{31}$$

$$= 0.9 * 0.2 * 0.1 * 0.8 * 0.8 * 0.2 * 0.9$$

$$= 0.0020736$$

Output probability for $s_1 s_2 s_3 s_3$

$$= p(1/s_1) * p(s_2/s_1) * p(3/s_2) * p(s_3/s_2) * p(2/s_3) \\ * p(s_3/s_3) * p(1/s_3)$$

$$= b_{11} * 0.2 * b_{23} * 0.2 * b_{32} * 1 * b_{31}$$

$$= 0.9 * 0.2 * 0.1 * 0.2 * 0.1 * 1 * 0.9$$

$$= 0.000324$$

$$\text{Total output probability for A} = 0 + 0.0020736 \\ + 0.000324$$

$$= 0.0023976$$

HMM for character B:

Output probability for character B:

output probability $s_1 s_1 s_2 s_3$

$$= p(1/s_1) * p(s_1/s_1) * p(3/s_1) * p(s_2/s_1) * p(2/s_2) \\ * p(s_3/s_2) * p(1/s_3)$$

$$= b_{11} * 0.8 * b_{13} * 0.2 * b_{22} * 0.2 * b_{31}$$

$$= 0.9 * 0.8 * 0 * 0.2 * 0.2 * 0.2 * 0.6$$

$$= 0$$

Output probability $s_1 s_2 s_2 s_3$

$$= P(1/s_1) * P(s_2/s_1) * P(3/s_2) * P(s_2/s_2) \\ * (2/s_2) * P(s_3/s_2) * P(1/s_3)$$

$$= b_{11} * 0.2 * b_{23} * 0.8 * b_{22} * 0.2 * b_{31}$$

$$= 0.9 * 0.2 * 0.8 * 0.8 * 0.2 * 0.2 * 0.6$$

$$= 0.0027648$$

Output probability $s_1 s_2 s_3 s_3$

$$= P(1/s_1) * P(s_2/s_1) * P(3/s_2) * P(s_3/s_2) * P(2/s_3) * P(s_3/s_3) \\ * P(1/s_3)$$

$$= b_{11} * 0.2 * b_{23} * 0.2 * b_{32} * 1 * b_{31}$$

$$= 0.9 * 0.2 * 0.8 * 0.2 * 0.4 * 1 * 0.6$$

$$= 0.006912$$

$$\text{Total probability for } B = 0 + 0.0027648 \\ + 0.006912 \\ = 0.0096768$$

Therefore character 'B' is more likely for
given observation sequence $\{1, 3, 2, 1\}$

Ans.